

## Game Theory

|                                                              |              |
|--------------------------------------------------------------|--------------|
| Section Id :                                                 | 64065359416  |
| Section Number :                                             | 8            |
| Section type :                                               | Online       |
| Mandatory or Optional :                                      | Mandatory    |
| Number of Questions :                                        | 6            |
| Number of Questions to be attempted :                        | 6            |
| Section Marks :                                              | 25           |
| Display Number Panel :                                       | Yes          |
| Section Negative Marks :                                     | 0            |
| Group All Questions :                                        | No           |
| Enable Mark as Answered Mark for Review and Clear Response : | No           |
| Section Maximum Duration :                                   | 0            |
| Section Minimum Duration :                                   | 0            |
| Section Time In :                                            | Minutes      |
| Maximum Instruction Time :                                   | 0            |
| Sub-Section Number :                                         | 1            |
| Sub-Section Id :                                             | 640653124031 |
| Question Shuffling Allowed :                                 | No           |

Question Number : 148 Question Id : 640653829417 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

**THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : GAME THEORY (COMPUTER BASED EXAM)"**

**ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?**

**CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.**

**(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)**

Options :

6406532786044. ✓ YES

6406532786045. ✗ NO

|                              |              |
|------------------------------|--------------|
| Sub-Section Number :         | 2            |
| Sub-Section Id :             | 640653124032 |
| Question Shuffling Allowed : | No           |

Question Id : 640653829418 Question Type : COMPREHENSION Sub Question Shuffling

**Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix**

**Question Numbers : (149 to 150)**

Question Label : Comprehension

An item is up for auction. Player 1 values the item at 3 while player 2 values the item at 5. Each player can bid either 0, 1 or 2. If player  $i$  bids more than player  $j$  then  $i$  wins the good and pays his bid, while the loser does not pay. If both players bid the same amount, then a coin is tossed to determine who the winner is, who gets the good and pays his bid while the loser pays nothing. Formulate the game in matrix form and answer the given subquestions.

**Sub questions**

**Question Number : 149 Question Id : 640653829419 Question Type : MCQ**

**Correct Marks : 2**

Question Label : Multiple Choice Question

Which player does not have strictly dominated strategy?

**Options :**

6406532786046. ✓ Player 1

6406532786047. ✗ Player 2

6406532786048. ✗ None

**Question Number : 150 Question Id : 640653829420 Question Type : MCQ**

**Correct Marks : 3**

Question Label : Multiple Choice Question

Which strategies survive Iterated Elimination of Strictly Dominated Strategy?

**Options :**

6406532786049. ✗ (0, 0)

6406532786050. ✗ (1, 1)

6406532786051. ✓ (2, 2)

6406532786052. ✗ None

**Sub-Section Number :**

3

**Sub-Section Id :**

640653124033

**Question Shuffling Allowed :**

Yes

**Question Number : 151 Question Id : 640653829421 Question Type : MSQ**

**Correct Marks : 3 Max. Selectable Options : 0**

Question Label : Multiple Select Question

(Hawk-Dove) The following game has been widely used in evolutionary biology to understand how “fighting” and “display” strategies by animals could coexist in a population. For a typical Hawk-Dove game there are resources to be gained (i.e. food, mates, territories, etc.) denoted as  $v$ . Each of two players can choose to be aggressive, called “Hawk” (H), or can be compromising, called “Dove” (D). If both players choose H then they split the resources, but lose some payoff from injuries, denoted as  $k$ . Assume that  $k > v/2$ . If both choose D then they split the resources, but engage in some display of power that a display cost  $d$ , with  $d < v/2$ . Finally, if player  $i$  chooses H

while  $j$  chooses D, then  $i$  gets all the resources while  $j$  leaves with no benefits and no costs. Describe the game as a game matrix and answer the following question.

The outcomes cannot be supported as pure strategy Nash equilibrium, given  $v = 10$ ,  $k = 6$ , and  $d = 4$ , is (are)

Options :

- 6406532786053. ✓ (H, H)
- 6406532786054. ✗ (H, D)
- 6406532786055. ✗ (D, H)
- 6406532786056. ✓ (D, D)

Sub-Section Number : 4  
Sub-Section Id : 640653124034  
Question Shuffling Allowed : No

Question Id : 640653829422 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (152 to 155)

Question Label : Comprehension

Determined to crack down on the drug trade, a city mayor puts more officers out on patrol to disrupt the business of drug dealers. A drug dealer in a neighborhood can work his trade either on a street corner or in the park. Each day, he decides where to set up shop, knowing that word about his location will travel among users. Because a good snitch is lacking, word does not travel to the police. The police officer on the beat then needs to decide whether she will patrol the park or the street corner, while not knowing where the drug dealer is hanging out that day. The decision of the officer and the dealer determine the extent of drug trades that day. Let suppose the total number of potential trades in the market is 100. A dealer's payoff is determined by the number of trades he consummates, while the officer's payoff is based on the number of trades she disrupts. The table below illustrates the payoffs for both parties according to their respective strategies in the game.

|                |               | Drug Dealer   |        |
|----------------|---------------|---------------|--------|
| Police Officer |               | Street Corner | Park   |
|                | Street Corner | 80, 20        | 0, 100 |
|                | Park          | 10, 90        | 60, 40 |

Based on the above data answer the given subquestions.

Sub questions

Question Number : 152 Question Id : 640653829423 Question Type : MCQ  
Correct Marks : 1  
Question Label : Multiple Choice Question

Choose the correct statement

**Options :**

6406532786057. ✖ There is a unique Nash equilibrium in pure strategy
6406532786058. ✖ There are finitely many Nash equilibria in this game in pure strategy
6406532786059. ✔ There is no Nash equilibrium in this game in pure strategy
6406532786060. ✖ None of these

**Question Number : 153 Question Id : 640653829424 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

Choose the correct statement

**Options :**

6406532786061. ✔ There is a unique Nash equilibrium in this game
6406532786062. ✖ There are finitely many Nash equilibria in this game
6406532786063. ✖ There is no Nash equilibrium in this game
6406532786064. ✖ None of these

**Question Number : 154 Question Id : 640653829425 Question Type : MCQ**

**Correct Marks : 2**

Question Label : Multiple Choice Question

What are the randomization probabilities of a police officer for patrolling the streets and patrolling the park respectively under equilibrium?

**Options :**

6406532786065. ✖  $6/13, 7/13$
6406532786066. ✖  $7/13, 6/13$
6406532786067. ✔  $5/13, 8/13$
6406532786068. ✖  $8/13, 5/13$

**Question Number : 155 Question Id : 640653829426 Question Type : MCQ**

**Correct Marks : 2**

Question Label : Multiple Choice Question

The randomization probabilities of the drug dealer choosing between street and park respectively under equilibrium are

**Options :**

6406532786069. ✖  $7/13, 6/13$
6406532786070. ✔  $6/13, 7/13$
6406532786071. ✖  $4/13, 9/13$
6406532786072. ✖  $9/13, 4/13$

**Sub-Section Number :**

5

**Sub-Section Id :**

640653124035

**Question Shuffling Allowed :**

No

**Question Id : 640653829427 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (156 to 163)**

Question Label : Comprehension

Consider a two-player game in which player 1 can choose A or B. The game ends if he chooses A while it continues to player 2 if he chooses B. Player 2 can either choose C or D, with the game ending after C and continuing again with player 1 after D. Player 1 then can choose E or F, and then the game ends after each of these choices.

Imagine that the payoffs following choice A by player 1 are (2, 0), following C by player 2 are (3, 1), following E by player 1 are (0, 0) and following F by player 1 are (1, 2)

Model this as an extensive form game and answer the given subquestions:

### Sub questions

**Question Number : 156 Question Id : 640653829428 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

This is a game of

**Options :**

6406532786073. ✓ Perfect information

6406532786074. ✗ Imperfect information

6406532786075. ✗ None

6406532786076. ✗ Can't be determined

**Question Number : 157 Question Id : 640653829429 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

How many terminal nodes does this game have?

**Options :**

6406532786077. ✗ 3

6406532786078. ✗ 2

6406532786079. ✓ 4

6406532786080. ✗ 5

**Question Number : 158 Question Id : 640653829430 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

How many information sets does this game have?

**Options :**

6406532786081. ✗ 2

6406532786082. ✓ 3

6406532786083. ✖ 4

6406532786084. ✖ 1

**Question Number : 159 Question Id : 640653829431 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

How many pure strategies does player 1 have?

**Options :**

6406532786085. ✖ 1

6406532786086. ✖ 2

6406532786087. ✖ 3

6406532786088. ✔ 4

**Question Number : 160 Question Id : 640653829432 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

How many pure strategies does player 2 have?

**Options :**

6406532786089. ✖ 1

6406532786090. ✔ 2

6406532786091. ✖ 3

6406532786092. ✖ 4

**Question Number : 161 Question Id : 640653829433 Question Type : MCQ**

**Correct Marks : 1**

Question Label : Multiple Choice Question

How many pure strategy Nash equilibria does this game have?

**Options :**

6406532786093. ✖ 1

6406532786094. ✖ 2

6406532786095. ✔ 3

6406532786096. ✖ 4

**Question Number : 162 Question Id : 640653829434 Question Type : MCQ**

**Correct Marks : 2**

Question Label : Multiple Choice Question

Which one of the Nash equilibria is also a subgame perfect equilibrium?

**Options :**

6406532786097. ✖ (AE, D)

6406532786098. ✔ (AF, D)